

as potent catalysts and could be used to render pollutants harmless. Nanosize iron, for example, can detoxify organic solvents, such as trichloroethylene. Other nanoparticles that are bioactive, such as silver and magnesium oxide, can kill bacteria and might be used in place of chlorine to disinfect water.

Nano-engineered membranes and filtration devices could be used to detect and remove viruses and other pollutants that are difficult to trap using current technologies. For instance, a preliminary technique employs imprinted polymer nanospheres to detect pharmaceuticals—a kind of pollution coming from households that is difficult to spot in waterways and was only recently discovered. Such nanoscale sensors might be helpful for real-time monitoring of these pollutants—everything from chemicals in Prozac to hormones in birth control pills—at water-treatment plants and industrial sites. Eventually, “smart” membranes with specifically tailored nanopores might be designed to both detect and remove such pollutants. With greater ability to filter out unwanted materials, industrial wastewater— and even the ocean—could become available to boost the supply of clean water.

The short term goals for Nanotechnology based clean water technologies would be the development of new techniques for remediation of water pollution. In another decade or so, nanotechnology is expected to have an impact on water treatment, beginning with nano absorbents and bioactive nanoparticles that could be integrated into existing purification systems. Before being used for water treatment, these nanomaterials would need to be evaluated for safety, including examining their toxicity, their transport, and their fate in the environment. Some of these visions are outlined below:

- 1. Desalination** - One way to expand the availability of drinking water in coastal communities is to turn to the sea and developing technologies for removing salt from seawater. The trouble with desalination is that it currently requires a lot of energy, so it is costly and all but guaranteed to grow more expensive in the future. There is a critical need to design alternative desalination methods that are more efficient. The aim is to develop smart membranes with antimicrobial surfaces and embedded sensors that can automatically adjust membrane performance. Nanotechnology is likely to play an important role in meeting that challenge.



Desalination of sea water